

Postcards from the future

PREDICTING THE FUTURE OF TECHNOLOGY

has always been a ridiculous exercise. The best examples of exactly how over the top it can get, are probably the covers of pulp science fiction books, as well as European postcards from the future. We do not have many of the things yet, such as humongous genetically engineered fruits, entire cities shielded from the elements, weather control devices, machines that digitally feed books to your brains, or large trains with entire apartment complexes on them. However, what is exciting is that over the course of the next decade, we are actually going to realise many of these dreams.

One of the best elements of science fiction and alternate history novel covers, often ambiently in the background, are all the airships. Believe it or not, there is actually a need for slow moving, steady aircraft. Even now, these are being used for prospecting, tourism and research. The Zeppelins are being operated for all these purposes already. The advent of automation, has blown new air into these vehicles. The Airlander 50 by Hybrid Air Vehicles, is being planned as a low cost freight craft. The airship is expected to be more economical than any air or land based alternatives, can deliver cargo to remote locations because of the lack of need of airport infrastructure, and can lift 50 tonnes of cargo. Automation will allow us to realise another cherished dream of science fiction – flying cars. Uber and Volocopter both have concepts for autonomous quadcopters ferrying passengers over short distances, to speed up travel. Dubai started the testing phase of its Autonomous Air Taxi service in 2017 itself, which is expected to last for another four years. A Boston based company, Terrafugia hopes to launch the TF-X autonomous flying car within the next decade, and it can fit into a regular garage.

Even a hundred years ago, people were predicting that the cities of the future will run on clean energy, without smoke or loud sounds polluting the environment. At this point of time, our cities are more polluted than ever before. Fortunately, there are two things in the near future that will definitely help clear up the environment. We are near to, or have crossed the tipping point, where any further degradation of the atmos-

phere will permanently and drastically affect the future of the planet. It has become of paramount importance to start taking pollutants out of the air, and stop dumping even more into it. Around the world, there are plans to deploy air filtration systems and drone swarms to remove carbon dioxide from the atmosphere and the seas. Over the course of the next decade, many countries are planning to shift to cleaner forms of energy production, and phase out coal-based power plants at the same time. The realisation of fusion power will be a big step, but solar panels and solar energy are also expected to get progressively cheaper. Electric vehicles are expected to increase exponentially. India plans to move away from fossil fuel vehicles completely, by using replaceable batteries for everything from trucks to cars to tractors. These batteries can be charged in places that now serve as fuel stations. Carpooling apps, ridesharing services, and subscription based vehicle ownership models mean less than half of the world's population will own a private vehicle after the next ten years.

When humanity was about to conquer the skies, the seas were a domain of equal importance. When the first aeronauts were imagined, of equal importance were the aquanauts. The sea was just another realm that man would eventually master. People had imagined underwater tours, races, and even humans riding on aquatic animals. None of this has happened. We have plans and timelines for establishing permanent settlements on Mars, but no concrete plans for a permanent underwater human settlement on Earth. I really cannot understand why this is the case, and hope to see some progress on this front over the next decade.

Shimizu Corporation, an architecture firm from Japan has conceptualised an "Ocean Spiral". At the top of the city is a 500m diameter sphere that serves as the base camp and supports the rest of the structure. Below the sphere, are three ballast balls to control the buoyancy. Originating from the Blue Garden, there is a long spiral to the sea floor, which houses the roads, utilities, offices and houses. Touching the sea floor, is an "Earth Factory", to store and reuse carbon dioxide, and to farm resources. The Ocean Spiral is a beautiful concept. We need an Atlantis already. 



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